Smart Research Intelligent System

Problem Statement No. 1 – Research Agent

The Challenge

Research activities often require significant time and effort to search for academic papers, review literature, extract key insights, and organize information from multiple sources.

Researchers, students, and professionals face challenges in identifying relevant research materials, understanding complex topics, and generating structured reports efficiently. The rapidly growing volume of scientific publications makes literature review and knowledge discovery increasingly difficult.

Manual research processes are time-consuming, repetitive, and may result in missed information or reduced productivity.

The Objective

Develop an AI-powered Research Agent that assists users throughout the research process by automating information retrieval, summarization, knowledge extraction, and report generation.

The system should:

- Search and retrieve relevant research information.
- Summarize academic papers and technical documents.
- Extract key findings and insights.
- Organize research knowledge effectively.
- Generate structured research reports.
- Improve research productivity and decision-making.
- Proposed Solution

The proposed solution, Smart Research Intelligent System, is an AI-powered Research Agent built using LangFlow, IBM watsonx.ai, and IBM Granite Models.

The system enables users to submit research queries through a conversational interface. IBM Granite models analyze the query, understand the research intent, and generate meaningful summaries and insights.

Technology Stack

IBM watsonx.ai

IBM Granite Model

LangFlow

IBM Cloud Lite

Agentic AI

Natural Language Processing (NLP)

Expected Outcome

The Smart Research Intelligent System acts as an intelligent digital research assistant capable of providing accurate research insights, generating summaries, extracting knowledge, and supporting users in conducting efficient and effective research.